

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250274013

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

Yang; Xiaofeng et al.

ELECTRIC MOTOR WITH AIRGAP AND STATOR SLOT COOLING

Abstract

An electric motor includes a stator having windings disposed within winding slots. The stator includes a radially inner stator core surface, a radially outer stator surface, and at least one stator passage extending radially a first distance from the inner stator core surface. A rotor is mounted inside the stator and includes at least one fluid passage extending radially through the rotor to the radially outer rotor surface. A fluid circulation arrangement is configured to receive a first liquid via a first passage and a gas via at least one second passage, and direct the first liquid and the gas, via centrifugal force, into the airgap and the at least one stator passage as the rotor rotates inside the stator to discharge the first liquid and the gas out through the airgap and the at least one stator passage radially, and through the windings axially, thereby cooling the electric motor.

Inventors: Yang; Xiaofeng (Troy, MI), Fatemi; Alireza (Canton, MI), Lahr; Derek F. (Ann Arbor, MI), Peng; Peng (Columbus, OH)

Applicant: GM GLOBAL TECHNOLOGY OPERATIONS LLC (Detroit, MI)

Family ID: 1000007723254

Assignee: GM GLOBAL TECHNOLOGY OPERATIONS LLC (Detroit, MI)

Appl. No.: 18/588575

Filed: February 27, 2024

Publication Classification

Int. Cl.: H02K9/19 (20060101); H02K1/20 (20060101); H02K1/32 (20060101)

U.S. Cl.:

CPC H02K9/19 (20130101); H02K1/20 (20130101); H02K1/32 (20130101);

Background/Summary

INTRODUCTION

[0001] The disclosure relates to an electric motor with airgap and stator slot cooling.

[0002] An electric motor is a machine that converts electric energy, i.e., electrical current into mechanical energy, i.e., mechanical torque. Electric motors are classified into two categories based on the direction of the magnetic field, i.e., axial flux motors and radial flux motors, each of which may be configured as an alternating current (AC) or a direct current (DC) type.

[0003] Radial flux motors include a rotor mounted within a stator having winding slots including windings. Each winding is formed by connecting a plurality of coils in series. The electric motor operates to generate torque based on an electromagnetic interaction between permanent magnets and a magnetic field created when coils within the electric motor are selectively energized.

[0004] As a byproduct of torque generation, electric motors produce thermal energy that may adversely affect motor performance and reliability. Cooling of the electric motor may reduce thermal stress seen, for example, by the rotor, stator, motor poles, windings and/or end-turns, providing longer motor life under or close to peak load. Additionally, electric motor cooling may generally quiet motor operation and enhance motor operation at higher speeds, as well as facilitate reduced motor inertia and packaging.

SUMMARY

[0005] An electric motor including a stator having windings disposed within winding slots, a rotor mounted inside the stator, and a fluid circulation arrangement is disclosed. The stator may include a radially inner stator core surface and a radially outer stator surface. The stator may further include at least one stator passage extending radially a first distance from the inner stator core surface.

[0006] The rotor mounted inside the stator defines a rotational axis, and may include axially opposite rotor ends and a radially outer rotor surface positioned proximate the radially inner stator surface, thereby establishing an airgap therebetween. The rotor may further include at least one fluid passage extending radially through the rotor to the radially outer rotor surface.

[0007] The fluid circulation arrangement may be configured to receive a first liquid via a first passage and a gas via at least one second passage, and direct the first liquid and the gas, via centrifugal force, into the airgap and the at least one stator passage as the rotor rotates inside the stator to discharge the first liquid and the gas out through the airgap and the at least one stator passage radially, and through the windings axially, thereby cooling the electric motor.

[0008] The at least one stator passage may extend a second distance from the inner stator core surface to an opening in the radially outer stator surface, the second distance greater than the first distance.

[0009] The fluid circulation arrangement may be configured to receive a second liquid through the opening in the radially outer stator surface via the at least one stator passage. The fluid circulation arrangement may further include a rotor shaft positioned coaxially with the rotor, fixed to the rotor, and may define the first shaft passage.

[0010] According to one aspect of the disclosure, the rotor shaft may define the at least one second passage.

[0011] According to another aspect of the disclosure, the first passage may include the stator passage, and the rotor may define the at least one second passage.

[0012] According to another aspect of the disclosure, the first passage may include the stator passage, and the at least one second passage may be disposed between the rotor, and a rotor shaft may be positioned coaxially with the rotor and fixed to the rotor.

[0013] The first passage may include the stator passage, and the rotor shaft defines the at least one second passage.

[0014] A stator ring may be disposed in the stator passage.

[0015] The stator may further include stator teeth that extend radially from the inner stator core surface and alternate with the winding slots circumferentially along a channel portion of the stator. The stator teeth within the stator passage may have a tapered configuration.

[0016] The stator teeth within the stator passage may be narrower than the stator teeth not within the stator passage.

[0017] The stator may further include stator teeth that extend radially from the inner stator core surface and alternate with the winding slots circumferentially along a channel portion of the stator. The stator teeth within the stator passage may be removed.

[0018] According to another aspect of the disclosure, an electric motor may include a stator including windings disposed within winding slots, a rotor mounted inside the stator, and a fluid circulation arrangement.

[0019] The stator may include a radially inner stator core surface and a radially outer stator surface. The stator may further include at least one stator passage extending radially a first distance from the inner stator core surface.

[0020] The rotor mounted inside the stator, defining a rotational axis, may include axially opposite rotor ends and a radially outer rotor surface positioned proximate the radially inner stator surface, thereby establishing an airgap therebetween. The rotor may further include at least one fluid passage extending radially through the rotor to the radially outer rotor surface.

[0021] The fluid circulation arrangement may be configured to receive a first liquid via a first passage and a gas via at least one second passage, and direct the first liquid and the gas, via centrifugal force, into the airgap and the at least one stator passage as the rotor rotates inside the stator to discharge the first liquid and the gas out through the airgap and the at least one stator passage radially, and through the windings axially, thereby cooling the electric motor.

[0022] The fluid circulation arrangement may further include a rotor impeller defining the plurality of fluid channels, wherein the radially outer rotor surface may define circumferentially distributed apertures fluidly connected to the respective plurality of fluid channels.

[0023] The rotor impeller may be arranged along the rotational axis centrally within the rotor.

[0024] The rotor may be a three-piece structure, and may include a first lateral rotor portion and a second lateral rotor, wherein the rotor impeller is may be disposed adjacent to both the first lateral rotor portion and the second lateral rotor portion.

[0025] A motor vehicle having such an electric motor as described above is also disclosed.

[0026] The above features and advantages, and other features and advantages of the present disclosure, will be readily apparent from the following detailed description of the embodiment(s) and best mode(s) for carrying out the described disclosure when taken in connection with the accompanying drawings and appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] FIG. 1 is a schematic illustration of a motor vehicle having a powertrain employing an electric motor-generator for propulsion.

[0028] FIG. 2 is a schematic partial cut-away perspective view of an electric motor-generator including a fluid circulation arrangement, according to one aspect of the disclosure.

[0029] FIG. 3 is a schematic cross-sectional front view illustration of an electric-motor generator, according to the disclosure.

[0030] FIG. 4 is a schematic cross-sectional side view illustration of an electric-motor generator, according to the disclosure.

[0031] FIG. 5 is a schematic cut-away sectional illustration of an electric-motor generator including

a stator passage, according to one aspect of the disclosure.

[0032] FIG. **6** is a schematic cut-away sectional illustration of an electric-motor generator including another stator passage, according to one aspect of the disclosure.

[0033] FIG. **7A** is a schematic cut-away section illustration of an electric-motor generator, according to another aspect of the disclosure.

[0034] FIG. **7B** is a schematic cut-away section illustration of an electric-motor generator, according to another aspect of the disclosure.

[0035] FIG. **8** is a schematic cut-away section illustration of an electric-motor generator including a stator ring, according to another aspect of the disclosure.

[0036] FIG. **9A** is an isometric sectional view of a stator, according to one aspect of the disclosure.

[0037] FIG. **9B** is a front view of the section of the stator in FIG. **9A**, according to one aspect of the disclosure.

[0038] FIG. **9C** is a schematic illustration of the stator in FIG. **9A**, according to another aspect of the disclosure.

[0039] FIG. **9D** is a schematic illustration of the stator in FIG. **9A**, according to another aspect of the disclosure.

[0040] FIG. **9E** is a schematic illustration of the stator in FIG. **9A**, according to another aspect of the disclosure.

[0041] FIG. **10** is a schematic illustration of a stator including a slot liner, according to the disclosure.

DETAILED DESCRIPTION

[0042] Embodiments of the present disclosure as described herein are intended to serve as examples. Other embodiments may take various and alternative forms. Additionally, the drawings are generally schematic and not necessarily to scale. Some features may be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present disclosure.

[0043] Certain terminology may be used in the following description for the purpose of reference only, and thus are not intended to be limiting. For example, terms such as “above” and “below” refer to directions in the drawings to which reference is made. Terms such as “front”, “back”, “fore”, “aft”, “left”, “right”, “rear”, “side”, “upward”, “downward”, “top”, and “bottom”, etc., describe the orientation and/or location of portions of the components or elements within a consistent but arbitrary frame of reference, which is made clear by reference to the text and the associated drawings describing the components or elements under discussion.

[0044] Furthermore, terms such as “first”, “second”, “third”, and so on may be used to describe separate components. Such terminology may include the words specifically mentioned above, derivatives thereof, and words of similar import, and are used descriptively for the figures, and do not represent limitations on the scope of the disclosure, as defined by the appended claims. Moreover, the teachings may be described herein in terms of functional and/or logical block components and/or various processing steps. It should be realized that such block components may include a number of hardware, software, and/or firmware components configured to perform the specified functions.

[0045] Referring to FIG. **1**, a motor vehicle **10** having a powertrain **12** is illustrated. The motor vehicle **10** may include, but not be limited to, a commercial vehicle, industrial vehicle, passenger vehicle, aircraft, watercraft, train or the like. It is also contemplated that the motor vehicle **10** may be a mobile platform, such as an airplane, all-terrain vehicle (ATV), boat, personal movement apparatus, robot, and the like to accomplish the purposes of this disclosure. The powertrain **12** includes a first power-source **14** depicted as an electric motor-generator and configured to generate a first power-source torque T.sub.1 for propulsion of the motor vehicle **10** via driven wheels **16** relative to a road surface.

[0046] The electric motor-generator **14** may be configured as a radial flux electric motor, wherein the magnetic flux is generated perpendicular to the motor's axis of rotation and an airgap between the machine's rotor and stator is arranged concentrically with the rotational axis.

[0047] As further illustrated in FIG. **1**, the powertrain **12** may also include a second power-source **20**, such as an internal combustion engine configured to generate a second power-source torque T.sub.2. The power-sources **14** and **20** may act individually and/or in concert to power the motor vehicle **10** and be operatively connected to a transmission assembly **22**.

[0048] The transmission assembly **22** may be configured to transmit first and/or second power-source torques T.sub.1, T.sub.2 to a final drive unit **24**, which in turn may be connected to the driven wheels **16**. The first power-source **14**, which for the remainder of the present disclosure will be referred to as an electric motor-generator, may be, for example, mounted to the second power-source **20**, mounted to (or incorporated into) the transmission assembly **22**, mounted to the final drive unit **24**, or be a stand-alone assembly mounted to the structure of the vehicle **10**.

[0049] As illustrated, the motor vehicle **10** further includes a programmable electronic controller **26** configured to communicate via a high-voltage BUS **27** and control the powertrain **12** to generate a predetermined amount of power-source torque (sum of T.sub.1 and T.sub.2), and various other vehicle systems. The motor vehicle **10** further includes a rechargeable energy storage system (RESS) **28**, such as one or more batteries, configured to generate and store electrical energy for powering the power-sources **14** and **20**.

[0050] Referring now to FIG. **2**, a general cross-section of the electric motor-generator **14** includes a rotationally fixed stator **30**, a rotor **36**, and a rotor shaft **52**. The stator **30** has a generally cylindrical core **32** including stator teeth **33**, and winding slots **34**. The stator core **32** has a radially inner stator core surface **32A**, a radially outer stator surface **32B**, and includes stator teeth **33** and winding slots **34** both of which extend radially outward from and alternate along the radially inner stator core surface **32A**.

[0051] The rotor **36** defines a rotational axis X and is mounted for rotation inside the stator **30**. The stator **30** may include multiphase AC windings **34A** arranged within the winding slots **34**, wherein the windings receive multiphase AC from a power inverter (not shown) to establish a rotating magnetic field exerting torque upon the rotor **36**. The stator windings **34A** are generally contained within the winding slots **34** with end turns **34C** of the windings **34A** extending beyond the limits of the cylindrical core **32** at axially opposite stator ends—a first end **32-1** and a second end **32-2** (FIG. **3**).

[0052] The rotor **36** has a ferromagnetic rotor core **38**. The rotor core **38** has axially opposite rotor core ends—a first end **38-1** and a second end **38-2**. The rotor core **38** may be constructed from a relatively soft magnetic material, such as laminated silicon steel. In a permanent magnet machine, stacked rotor laminations may include voids forming interior pockets for carrying permanent magnets. In an induction machine, stacked laminations may include peripheral slots for carrying conduction bars. Alternative rotor constructions are also used in the industry and may include, for example, surface mounted permanent magnet and wire wound rotors. The rotor core **38** has a radially outer rotor surface **38A** positioned proximate the radially inner stator core surface **32A**, thereby establishing an airgap **40** therebetween (shown in FIGS. **2-4**). The airgap **40** extends between the axially opposite first end **38-1** and second end **38-2** of the rotor core.

[0053] As illustrated in FIGS. **2-4**, the rotor **36** further includes a fluid circulation arrangement **42** having one or more fluid channels **44**, each extending radially through the rotor to the radially outer rotor surface **38A**. The fluid circulation arrangement **42** is configured to receive a first liquid **46-1** via a first passage **50-1** and a gas **48** via one or more second passages **50-2** along the rotational axis X. Specifically, the first liquid **46-1** may be a pressurized oil and the gas **48** may be air. The fluid circulation arrangement **42** is further configured to direct the first liquid **46-1** and the gas **48**, via centrifugal force, into the airgap **40** as the rotor **36** rotates inside the stator **30** to discharge the first liquid **46-1** and the gas **48** out of the airgap **40** at the axially opposite ends **38-1**, **38-2** of rotor **36**.

The fluid circulation arrangement **42** is thereby configured to cool the electric motor-generator **14**. [0054] The rotor shaft **52** is positioned coaxially with the rotor **36**. The rotor shaft **52** may be fixed to the rotor **36** and define a passage **50** including the first passage **50-1** for supplying the liquid **46** to the fluid channel(s) **44**. Specifically, the first passage **50-1** may be arranged along the rotational axis X and be configured to receive pressurized first liquid **46-1** from one end of the rotor shaft **52**. From the first passage **50-1**, the first liquid **46-1** may flow through liquid inlets configured as one or more third passages **50-3** into the internal rotor fluid channels **44**. As shown in FIGS. 2, 3, and 5, the third passage(s) **50-3** may be cross-drilled holes arranged substantially perpendicular to the first passage **50-1**. As the radial flux motor-generator **14** generates the first power-source torque $T_{sub.1}$, the rotor shaft **52** spins about the axis of rotation X together with the rotor **36**. The rotor shaft **52** may be rotatably supported at both rotor core **38** ends (the first end **38-1** and the second end **38-2**) by bearings (not shown) and may transfer the first power-source torque $T_{sub.1}$ generated by the rotor **36** to the driven wheels **16** via the transmission assembly **22** (shown in FIG. 1).

[0055] The passage **50** further defines the second passage **50-2** for supplying the gas **48** to the fluid channel(s) **44**. Alternatively, the rotor **36** may define the second passage(s) **50-2** (not shown). In the illustrated example, the second passage(s) **50-2** may be arranged substantially parallel to the rotational axis X. The gas **48** may be drawn into the second passage **50-2** by the rotation of the rotor **36**, with a vacuum generated in the second passage resulting from the communication of the second passage with the radially disposed fluid channel(s) **44**. As illustrated in FIG. 2, the first and second passages **50-1**, **50-2** are in fluid communication with each of the fluid channels **44** in the fluid circulation arrangement **42**. The first liquid **46-1** and gas **48** may therefore be combined within the fluid channel(s) **44** thereby establishing a two-phase mixture subsequently injected into the airgap **40**.

[0056] According to one aspect of the disclosure, the fluid circulation arrangement **42** further includes a rotor impeller **54** having a plurality of blades **54A** defining a plurality of fluid channels **44**. The rotor impeller **54** is arranged perpendicular to the rotational axis X and fixed centrally within the rotor **36**. As shown in FIG. 2, the plurality of fluid channels **44** define a radially disposed channel pattern of the rotor impeller **54**. As illustrated in FIG. 3, the channels **44** extend through the rotor impeller **54** to the radially outer rotor surface **38A**. As such, the rotor **36** includes a three-piece structure (shown in FIGS. 2-4), including a first lateral rotor portion **36-1**, a second lateral rotor portion **36-2** and the rotor impeller **54** disposed adjacent to both the first lateral rotor portion **36-1** and the second lateral rotor portion **36-2**.

[0057] According to one aspect of the present disclosure, as illustrated in FIG. 5 with continued reference to FIGS. 2-4, the stator **30** further includes a stator passage **30-1** that extends radially outward from the radially inner stator core surface **32A** a first distance $D_{sub.1}$. The fluid circulation arrangement **42** is configured to receive the first liquid **46-1** via the first passage **50-1** and the gas **48** via the second passage **50-2**, and direct the first liquid **46-1** and the gas **48**, via centrifugal force, into the airgap **40** and the at least one stator passage **30-1** extending the first distance $D_{sub.1}$, as the rotor **36** rotates inside the stator **30** to discharge the first liquid **46-1** and the gas out through the airgap **40** and the at least one stator passage **30-1** radially, and through the windings **34A** axially, thereby cooling the electric motor **14**.

[0058] According to another aspect of the disclosure, as illustrated in FIG. 6 with continued reference to FIGS. 2-4, the stator **20** further includes a stator passage **30-1** that extends radially outward from the radially inner stator core surface **32A** a second distance $D_{sub.2}$ to an opening **35** in the radially outer stator surface **32B**. The fluid circulation arrangement **42** is configured to receive the first liquid **46-1** via the first passage **50-1** and the gas **48** via the second passage **50-2**, and a second liquid **46-2** via the stator passage **30-1** through the opening **35** in the radially outer stator surface **32B** of the stator **30**.

[0059] The fluid circulation arrangement **42** is configured to direct the first liquid **46-1**, the second liquid **46-2**, and the gas **48**, via centrifugal force, into the airgap **40** and the at least one stator

passage **30-1**, and further configured to direct the second liquid **46-2**, via centrifugal force, through the at least one stator passage **30-1** as the rotor **36** rotates inside the stator **30** to discharge the first liquid **46-1**, the second liquid **46-2**, and the gas out through the airgap **40** and the at least one stator passage **30-1** radially, and through the windings **34A** axially, thereby cooling the electric motor **14**. [0060] According to another aspect of the disclosure, as illustrated in FIG. 7A, the first passage **50-1** includes the stator passage **30-1**, and the rotor **36** defines the second passages **50-2**. i.e., the second passages **50-2** are disposed within the rotor **36**. One feature of this configuration is that the second passages **50-2** avoids the blockage of air flow during large oil flow rates such that the second passage **50-2** is for the flow of air and not oil.

[0061] According to another aspect of the disclosure, as illustrated in FIG. 7B, the first passage **50-1** includes the stator passage **30-1**, and the second passages **50-2** are disposed between the rotor **36** and the rotor shaft **52** positioned coaxially with the rotor **36** and fixed to the rotor **36**. One feature of this configuration is that the second passages **50-2** are closer to the rotor shaft **52** there will be a larger centrifugal force and pump more air into the airgap.

[0062] As illustrated in both FIGS. 7A and 7B, the first liquid **46-1**, for example, an oil, is pumped in through an oil inlet (opening **35**) in the stator **30** into the stator passage **30-1**, and the gas **48** is pumped in through air inlets **60**. As such, the gas **48** is separate from the liquid **46-1**.

[0063] According to another aspect of the disclosure, as illustrated in FIG. 8, the first passage **50-1** includes the stator passage **30-1**. The rotor shaft **52** defines the at least one second passage **50-2**. The first liquid **46-1**, for example but not limited to, oil is pumped into the stator passage **30-1**, while gas **48**, for example but not limited to, air is pumped into the at least one second passage **50-2**.

[0064] A non-flux carrying stator ring **66** is disposed within the stator passage **30-1**. The non-flux carrying materials could include paramagnetic or dielectrics, such as plastics, composites, polymers, or non-ferrous metals including aluminum. The stator ring **66**, which includes a feed hole **68**, distribution holes **70**, and an oil/air dam **72**, is operable to facilitate an even distribution of the oil within the electric motor **14**.

[0065] Referring now to FIGS. 9A and 9B, an isometric sectional view and a front view of a stator **30** having a ring portion **30A** and a channel portion **30B** are illustrated. The stator **30** includes stator teeth **33** that alternate with winding slots **34** circumferentially along the channel portion **30B**. The stator teeth **33** extend radially the ring portion **30A** toward an inner stator core surface **32A** a distance $D_{sub.1}$, and include tips **33A** located at the inner stator core surface **32A**. Windings **34A** are disposed within slot liners **34B**, which are disposed within the winding slots **34**.

[0066] Referring now to FIGS. 9C-9E, stator teeth **30** can be modified or removed within the stator passage **30-1** to promote oil flow from the rotor **36** into the winding slots **34**.

[0067] As illustrated in FIG. 9C, the tips **33A** of the stator teeth **33** within the stator passage **30-1** may be removed to provide additional room for the oil to flow into the winding slots **34**. In this example, the stator teeth **33** have a tapered configuration in which each of the stator teeth **33** located within the stator passage **30-1** taper in width from the ring portion **30A** to the inner stator core surface **32A**.

[0068] As illustrated in FIG. 9D, an average width W of the stator teeth **33** within the stator passage **30-1** may be reduced to provide additional room for the oil to flow into the winding slots **34**. In this configuration, the average width W of the stator teeth **33** within the stator passage **30-1** is up to 20% less than the stator teeth **33** not within the stator passage **30-1**.

[0069] As illustrated in FIG. 9E, the stator teeth **33** may be removed all together within the stator passage **30-1** to provide additional room for the oil to flow into the winding slots **34**.

[0070] Referring now to FIG. 10, the slot liner **34B** can include multiple different configurations of cutouts or openings through the slot liner **34B** depending on a level of oil desired to enter the slot liner **34B** for cooling the windings **34A**. In one example, slot liner **34(I)** includes a cutout that divides the slot liner **34B** (I) into axial halves. In another example, slot liner **34B** (II) includes a

single cutout along a radially inner edge and a single cutout along a radially outer edge with the single cutouts being located midway between axial ends of the slot liner **34B** (II). In yet another example, slot liner **34B** (III) includes a single cutout along a radially outer edge and multiple cutouts, such as five cutouts, along a radially inner edge.

[0071] The detailed description and the drawings or figures are supportive and descriptive of the disclosure, but the scope of the disclosure is defined solely by the claims. While some of the best modes and other embodiments for carrying out the claimed disclosure have been described in detail, various alternative designs and embodiments exist for practicing the disclosure defined in the appended claims. Furthermore, the embodiments shown in the drawings, or the characteristics of various embodiments mentioned in the present description are not necessarily to be understood as embodiments independent of each other. Rather, it is possible that each of the characteristics described in one of the examples of an embodiment may be combined with one or a plurality of other desired characteristics from other embodiments, resulting in other embodiments not described in words or by reference to the drawings. Accordingly, such other embodiments fall within the framework of the scope of the appended claims.

Claims

1. An electric motor comprising: a stator including windings disposed within winding slots, the stator having a radially inner stator core surface and a radially outer stator surface, wherein the stator includes at least one stator passage extending radially a first distance from the radially inner stator core surface; a rotor mounted inside the stator, defining a rotational axis, and having axially opposite rotor ends and a radially outer rotor surface positioned proximate the radially inner stator surface, thereby establishing an airgap therebetween, wherein the rotor includes at least one fluid passage extending radially through the rotor to the radially outer rotor surface; and a fluid circulation arrangement configured to receive a first liquid via a first passage and a gas via at least one second passage, and direct the first liquid and the gas, via centrifugal force, into the airgap and the at least one stator passage as the rotor rotates inside the stator to discharge the first liquid and the gas out through the airgap and the at least one stator passage radially, and through the windings axially, thereby cooling the electric motor.
2. The electric motor as recited in claim 1, wherein the at least one stator passage extends a second distance from the radially inner stator core surface to an opening in the radially outer stator surface, the second distance greater than the first distance.
3. The electric motor as recited in claim 2, wherein the fluid circulation arrangement is configured to receive a second liquid through the opening in the radially outer stator surface via the at least one stator passage.
4. The electric motor as recited in claim 3, wherein the fluid circulation arrangement further includes a rotor shaft positioned coaxially with the rotor, fixed to the rotor, and defining the first passage.
5. The electric motor as recited in claim 4, wherein the rotor shaft defines the at least one second passage.
6. The electric motor as recited in claim 2, wherein the first passage includes the stator passage, and the rotor defines the at least one second passage.
7. The electric motor as recited in claim 2, wherein the first passage includes the stator passage, and the at least one second passage is disposed between the rotor, and a rotor shaft positioned coaxially with the rotor and fixed to the rotor.
8. The electric motor as recited in claim 2, wherein the first passage includes the stator passage, and the rotor shaft defines the at least one second passage.
9. The electric motor as recited in claim 8, further including a stator ring disposed in the stator passage.

- 10.** The electric motor as recited in claim 1, wherein the stator further includes stator teeth that extend radially from the inner stator core surface and alternate with the winding slots circumferentially along a channel portion of the stator, and wherein the stator teeth within the stator passage have a tapered configuration.
- 11.** The electric motor as recited in claim 10, wherein the stator teeth within the stator passage are narrower than the stator teeth not within the stator passage.
- 12.** The electric motor as recited in claim 1, wherein the stator further includes stator teeth that extend radially from the inner stator core surface and alternate with the winding slots circumferentially along a channel portion of the stator, and wherein the stator teeth within the stator passage are removed.
- 13.** An electric motor comprising: a stator including windings disposed within winding slots, the stator having a radially inner stator core surface and a radially outer stator surface, wherein the stator includes at least one stator passage extending radially a first distance from the radially inner stator core surface; a rotor mounted inside the stator, defining a rotational axis, and having axially opposite rotor ends and a radially outer rotor surface positioned proximate the radially inner stator core surface, thereby establishing an airgap therebetween, wherein the rotor includes at least one fluid passage extending radially through the rotor to the radially outer rotor surface; and a fluid circulation arrangement configured to receive a first liquid via a first passage and a gas via at least one second passage, and direct the first liquid and the gas, via centrifugal force, into the airgap and the at least one stator passage as the rotor rotates inside the stator to discharge the first liquid and the gas out through the airgap and the at least one stator passage radially, and through the windings axially, thereby cooling the electric motor; wherein the fluid circulation arrangement further includes a rotor impeller defining a plurality of fluid channels, and wherein the radially outer rotor surface defines circumferentially distributed apertures fluidly connected to a respective plurality of fluid channels; the rotor impeller is arranged along the rotational axis centrally within the rotor; the rotor has a three-piece structure, including a first lateral rotor portion and a second lateral rotor portion, wherein the rotor impeller is disposed adjacent to both the first lateral rotor portion and the second lateral rotor portion.
- 14.** A motor vehicle comprising: an electric motor configured to generate torque for propulsion of the motor vehicle, the electric motor including: a stator including windings disposed within winding slots, the stator having a radially inner stator core surface and a radially outer stator surface, wherein the stator includes at least one stator passage extending radially a first distance from the radially inner stator core surface; a rotor mounted inside the stator, defining a rotational axis, and having axially opposite rotor ends and a radially outer rotor surface positioned proximate the radially inner stator core surface, thereby establishing an airgap therebetween, wherein the rotor includes at least one fluid passage extending radially through the rotor to the radially outer rotor surface; and a fluid circulation arrangement configured to receive a first liquid via a first passage and a gas via at least one second passage, and direct the first liquid and the gas, via centrifugal force, into the airgap and the at least one stator passage as the rotor rotates inside the stator to discharge the first liquid and the gas out through the airgap and the at least one stator passage radially, and through the windings axially, thereby cooling the electric motor.
- 15.** The motor vehicle as recited in claim 14, wherein the at least one stator passage extends a second distance from the radially inner stator core surface to an opening in the radially outer stator surface, the second distance greater than the first distance.
- 16.** The motor vehicle as recited in claim 15, wherein the fluid circulation arrangement is configured to receive a second liquid through the opening in the radially outer stator surface via the at least one stator passage.
- 17.** The motor vehicle as recited in claim 16, wherein the fluid circulation arrangement further includes a rotor shaft positioned coaxially with the rotor, fixed to the rotor, and defining the first passage.

- 18.** The motor vehicle as recited in claim 17, wherein the rotor shaft defines the at least one second passage.
- 19.** The motor vehicle as recited in claim 15, wherein the first passage includes the stator passage, and the rotor defines the at least one second passage.
- 20.** The motor vehicle as recited in claim 15, wherein the first passage includes the stator passage, and the at least one second passage is disposed between the rotor, and a rotor shaft positioned coaxially with the rotor and fixed to the rotor.
-